Cambridge Assessment International Education

Cambridge O Level

CHEMISTRY **5070/21**

Paper 2 Theory **October/November 2021**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	‘List rule’ guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Question	Answer	Marks
1(a)	sulfur dioxide	**1**
1(b)	iron(II) oxide	**1**
1(c)	aluminium oxide	**1**
1(d)	calcium oxide	**1**
1(e)	sulfur dioxide	**1**

Question	Answer	Marks
2(a)	21(%)	**1**
2(b)	carbon dioxide is an acidic oxide (1) reacts with base / reacts with alkali / neutralised by sodium hydroxide (1)	**2**
2(c)	air liquefied (1) fractional distillation (1)	**2**
2(d)	glowing splint (1) relights (1)	**2**
2(e)(i)	photochemical / redox	**1**
2(e)(ii)	lightning	**1**
2(f)	(more) skin cancer / (more) sunburn / (more) harm to eyes	**1**

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Question	Answer	Marks
3(a)(i)	6 minutes	**1**
3(a)(ii)	initial gradient steeper and starting from origin (1) line levels off at 34 cm^3 gas (1)	**2**
3(b)(i)	rate of reaction increases (no mark) **and** particles move faster / particles have more kinetic energy (1) more particles have activation energy (or above) / more successful collisions / more energetic collisions (1)	**2**
3(b)(ii)	rate of reaction decreases (no mark) **and** fewer particles exposed on surface / fewer particles (of iron) per unit area / fewer particles per cm^2 (1) collision frequency decreases / fewer particles collide per second / collision rate decreases / collisions less often (1)	**2**
3(c)	mol iron = $\frac{3.36}{56}$ **OR** 0.060 mol (1) volume of hydrogen = 1.44 (dm^3) (1)	**2**
3(d)	add (aqueous) potassium iodide (1) colourless to brown solution (1)	**2**

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Question	Answer	Marks
4(a)	breakdown of (long chain) alkanes / decomposition of (long chain) alkanes (1) into shorter chain (hydrocarbons) / by heat / thermally (1)	**2**
4(b)(i)	hydrogen	**1**
4(b)(ii)	addition	**1**
4(c)	$C_2H_4 + H_2O \rightarrow C_2H_5OH$	**1**
4(d)	C H O $\frac{54.5}{12}$ $\frac{9.10}{1}$ $\frac{36.4}{16}$ **OR** 4.54 9.10 2.275 (1) C_2H_4O (1)	**2**

Question	Answer	Marks
5(a)	(magnesium) loses electrons (1)	**1**
5(b)	reactants on the left and products on the right and reactant line above product line (1) arrow downwards between reactants and products with ΔH label (1)	**2**
5(c)	iodide	**1**
5(d)	layers made of ions (1) layers slide (1)	**2**
5(e)	more reactive the metal (in the carbonate) the harder it is to break down (the carbonate)	**1**

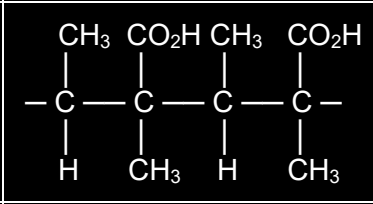
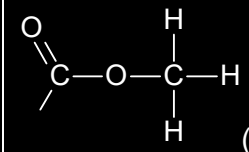
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Question	Answer	Marks
6(a)	bond breaking endothermic **AND** bond making exothermic / energy absorbed to break bonds **AND** energy released on making bonds (1) more energy released than absorbed (1)	**2**
6(b)(i)	iodine **AND** potassium chloride	**1**
6(b)(ii)	chlorine is more reactive than bromine / bromine is less reactive than chlorine	**1**
6(c)	$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$	**1**

Question	Answer	Marks
7(a)	**Any two from:** high melting point for Ag / high boiling point for Ag / **ORA** for K (1) high density for Ag / **ORA** for K (1) hard for Ag / soft for K (1)	**2**
7(b)	protons: **47** (1) neutrons: **62** (1) electrons: **46** (1)	**3**
7(c)	$2K + 2H_2O \rightarrow 2KOH + H_2$	**1**
7(d)(i)	equilibrium moves to the left / less product formed / more reactant formed (1) decreasing temperature pushes the reaction in the direction of releasing energy / decreasing temperature pushes the reaction in the direction of the exothermic reaction (1)	**2**

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Question	Answer	Marks
7(d)(ii)	equilibrium moves to the left / less product formed / more reactant formed (1) reaction goes in the direction to reduce the concentration of carbon dioxide (1)	**2**

Question	Answer	Marks
8(a)	has C=C bond / has a carbon-carbon double bond	**1**
8(b)	$C_5H_8O_2$	**1**
8(c)(i)	(solution which) contains hydrogen ions / hydrogen ion donor	**1**
8(c)(ii)	(acid which) is incompletely ionised / (acid which) is not completely dissociated	**1**
8(d)(i)	addition	**1**
8(d)(ii)	 (2 marks)	**2**
8(e)	completed with  (1)	**1**
8(f)	$2CH_3OH + 3O_2 \rightarrow 2CO_2 + 4H_2O$ correct formulae (1) balancing (1)	**2**

Question	Answer	Marks
9(a)	hydrogen: (cracking) hydrocarbons (1) nitrogen: air (1)	**2**
9(b)	$NaOH = \frac{4.5}{40}$ **OR** 0.1125 (mol) (1) $(NH_4)_2SO_4 = \frac{50}{1000} \times 1.25$ **OR** 0.0625 (mol) (1) ($(NH_4)_2SO_4$ because 0.0625×2) = 0.125 **OR** ($(NH_4)_2SO_4$ because $0.1125 \div 2$) = 0.056 (1)	**3**
9(c)	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$ / $4OH^- - 4e^- \rightarrow O_2 + 2H_2O$ (1)	**1**
9(d)	reduction	**1**
9(e)	**Any three from**: (nitrates) increase growth of algae / (nitrates) increase growth of water plants / algal bloom (1) plants (on surface) block sunlight and plants die (1) bacteria feed on dead plants and use up oxygen (1) (without oxygen) fish die / (without oxygen) water organisms die (1)	**3**

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Question	Answer	Marks
10(a)(i)	values between 5.40 and 11.00 (inclusive of these values)	**1**
10(a)(ii)	decrease down the group / increase up the group	**1**
10(a)(iii)	liquid **AND** 1600 (°C) is lower than the boiling point **AND** 1600 (°C) is higher than the melting point	**1**
10(b)(i)	**P** is a giant covalent structure / giant covalent lattice (1) **P** (all) bonds (throughout lattice) are strong (if covalent or bonds between atoms already mentioned) (1) **Q** is a simple molecular structure / small molecules / simple molecule (1) **Q** weak forces between molecules (1)	**4**
10(b)(ii)	4 pairs of bonding electrons between each C*l* and Si **AND** 6 unpaired electrons on each chlorine	**1**
10(c)	(molar mass of $Si(CH_3)_4$) = 88 (1) 54.5(%) (1)	**2**

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